

From jun-2018 | Page 26:

Q8(a)(ii): Calculate, in minutes, the time taken for this amount of charge to flow through the resistor.

Q8(a)(iii): Calculate the total energy transferred by the 6.0 V power supply when a charge of 42 C flows through the resistor.

From jun-2018 | Page 27:

Q8(b): Explain why the resistor becomes warm.

Q8(c): Deduce how the resistors have been arranged inside the cardboard tube.

From jun-2018 | Page 32:

Q10(a)(i): Calculate the potential difference across each lamp.

Q10(a)(ii): Calculate the resistance of each lamp.

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Q10(b): Explain, with the aid of a circuit diagram, the method a student could use to investigate how the resistance of a single lamp changes with the potential difference across the lamp.

From jun-2019 | Page 6:

Q3(b)(i): Calculate the value of resistance that is missing from the table.

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Q3(b)(ii): The student writes this conclusion: 'The resistance of the lamp is directly proportional to the potential difference.' Comment on the student's conclusion. Use information from Figure 5 in your answer.

Q3(b)(iii): Describe how the student could add a component to the circuit that would provide a continuously variable voltage across the lamp.

From jun-2019 | Page 28:

Q10(a)(i): Calculate the resistance of the immersion heater.

From jun-2020 | Page 26:

Q8(a)(i): Add to Figure 19: ■ a voltmeter to find the potential difference across resistor R ■ another resistor in parallel with resistor R.

Q8(a)(ii): State the measurements that the student must take to find the overall resistance of the resistors in parallel.

From jun-2020 | Page 27:

Q8(a)(iii): State the resistance of a single resistor.

Q8(a)(iv): Comment on the relationship between the overall resistance of the circuit and the number of resistors in parallel. Use information from Figure 20 to support your answer.

Q8(b)(i): Calculate the potential difference across the $15\ \Omega$ resistor.

From jun-2022 | Page 4:

Q2(a)(i): Calculate the resistance of the lamp.

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Q2(b): State and explain the difference between the brightness of the lamp in Figure 3 and the brightness of a lamp in Figure 4.

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Q2(c): Draw a diagram of the circuit that the student should use.

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Q8(b)(i): Describe how the resistance of this thermistor varies with temperature.

Q8(b)(ii): Draw the tangent to the curve at a temperature of $30\ ^\circ\text{C}$, to find the rate of change of resistance with temperature at $30\ ^\circ\text{C}$.

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Q8(c)(i): Explain one improvement in measurement that the student could make in the investigation.

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Q8(c)(ii): Explain why method 2 gives more precise results than method 1.

From jun-2023 | Page 4:

Q2(a): Add connecting wires, a voltmeter and an ammeter to complete the circuit in Figure 3 so that the students can determine the resistance of the piece of iron wire.

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Q2(b)(i): Give the name of one additional piece of apparatus the students would need.

Q2(b)(ii): Draw a straight line of best fit on Figure 4.

Q2(b)(iii): Use Figure 4 to estimate the resistance of a 100 cm length of the iron wire.

Q2(b)(iv): Explain how the variable resistor is used to prevent the iron wire from becoming too hot.

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Q2(c): Calculate the resistance of this piece of wire.

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Q8(b)(i): Comment on this claim. Use this equation to support your answer $t = E / (V \times I)$

Q8(b)(ii): Calculate the total charge that moves into the battery while it is being recharged. Use the equation $E = Q \times V$

From june-2024 | Page 5:

Q2(a): The current at P is

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Q2(b)(i): On Figure 5, draw a voltmeter connected to measure the voltage (p.d.) across the resistors.

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Q2(b)(ii): Using data from the table in Figure 6, predict the current from the power supply when there are 4 resistors in the circuit.

Q2(b)(iii): Using data from the table in Figure 6, calculate the resistance of only 1 resistor.

Q2(b)(iv): Using data from the table in Figure 6, explain what happens to the total resistance of the circuit as the number of resistors in parallel decreases.

From june-2024 | Page 25:

Q8(a): Complete the circuit diagram needed for this investigation.

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Q8(b)(i): Suggest one advantage of using a data logger instead of meters in this investigation.

Q8(b)(ii): Describe how current varies with potential difference in the graph in Figure 20.

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Q8(b)(iii): Use data from the graph in Figure 20 to show how the resistance changes with potential difference for the filament lamp.

Q8(d): Explain how, starting with the meaning of potential difference, we may obtain an equivalent unit of potential difference.

From nov-2021 | Page 6:

Q2(a)(i): Describe the relationship between the current and the voltage as shown in the graph in Figure 4.

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Q2(a)(ii): Use the values of the voltage and current at point P and at point Q on the graph in Figure 4 to complete the table in Figure 5.

Q2(a)(iii): Calculate the resistance of the filament lamp when the voltage is 4.5 V and the current is 51 mA.

Q2(a)(iv): Explain why the resistance of the filament lamp changes as the voltage across it increases.

From nov-2021 | Page 23:

Q8(a)(i): Draw on Figure 17 to show how the circuit should be completed so that the current in the circuit and voltage across the device can be measured.

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Q8(a)(iii): Calculate the energy transferred to the interior light in 7 minutes.

From nov-2021 | Page 25:

Q8(b)(i): Calculate the current in the wires to the battery when all the devices are switched on.

Q8(b)(ii): State how the overall resistance of the circuit changes when any one of the devices is switched off.

Q8(b)(iv): Explain why the wires to the battery in a car are thicker than the wires that connect each device to its switch and its fuse.

From specimen | Page 8:

Q3(a): Give three modifications the student should make to the circuit so that the circuit works correctly.

From specimen | Page 9:

Q3(b)(i): Give one reason for using the water bath.

Q3(b)(ii): State one way the student could develop this experimental procedure to investigate temperatures outside this range.

From specimen | Page 11:

Q3(c)(i): Compare and contrast how the resistances of component A and component B vary with temperature.

Q3(c)(ii): Which of these is the current in component A when the temperature is 80°C?

